

Heat Transfer

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Thermal energy moves from warmer to cooler areas by conduction through direct contact, convection through fluid movement, and radiation by electromagnetic waves.
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Round 1 - Retrieve It

1. What is conduction?

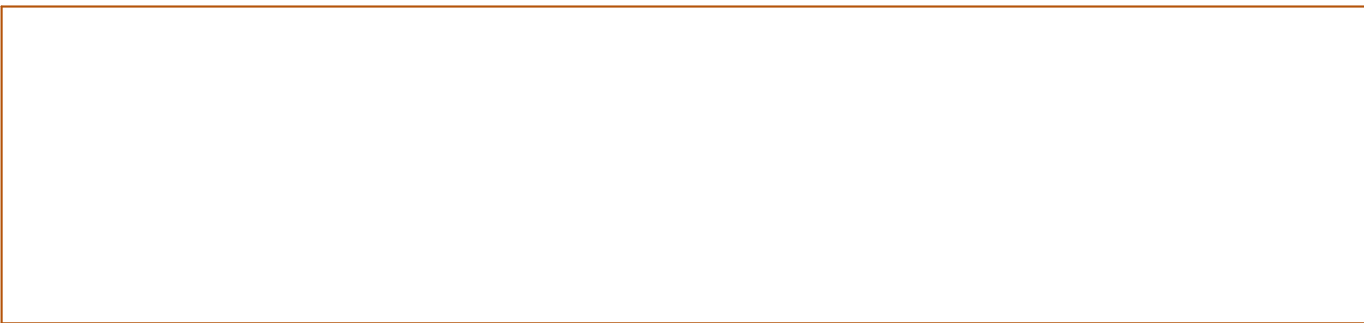
2. What is convection?

3. What is radiation?

4. In what overall direction does heat move spontaneously?

Round 2 - Sketch It

Draw a pot of water on a hot stove. Label one example each of conduction, convection, and radiation in or around the setup.



Heat Transfer - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. A metal spoon gets hot while sitting in soup. Main heat-transfer method through the spoon?

2. Warm water rises and cool water sinks in a pot. Main method?

3. You feel warmth from the Sun across space. Main method?

Round 4 - Reason from Evidence

At a campfire, you may experience more than one type of heat transfer. Describe two methods and the evidence for each.

Heat Transfer - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Heat Transfer in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect heat transfer to particle motion. How does warmer matter differ from cooler matter at the particle level?

E**Extend**

Design one change to a cup that would slow heat transfer from a hot drink, and explain which transfer method your design reduces.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Heat Transfer

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Heat transfer through direct contact or particle collisions, especially in solids.
2. Heat transfer by bulk movement/circulation in liquids or gases.
3. Energy transfer by electromagnetic waves; no matter is required between source and receiver.
4. From warmer to cooler areas.

Sketch It - look for

Conduction through pot/handle or contact; convection currents in water/air; radiation from burner/flame/hot surfaces.

Apply It

1. Conduction.
2. Convection.
3. Radiation.

Reason from Evidence - look for

Radiation from fire to body; convection as hot air/smoke rises and moves; conduction if touching a heated object. Need mechanism-based explanation.

ACE - Extend look-for

Accept insulation, lid, reflective surface, reduced conduction/convection/radiation with clear reasoning.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.